# NAM NGIEP 2A HYDROPOWER PROJECT

| | |
|---|---|
| Project title | Nam Ngiep 2A Hydro power project |
| Project ID | 4934 |
| Crediting period | 08-August-2024 to 07-August-2031 |
| Original date of issue | 02-April-2024 |
| Most recent date of issue | 02-April-2024 |
| Version | 01 |
| *VCS Standard* Version | 4.6 |
| Prepared by | Kosher Climate India Private Limited |

# CONTENTS

# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

Nam Ngiep 2A Hydropower project (hereafter referred to as "the project") is developed by Nonghai Group Sole Company Limited in Sanluang Town, Khoun District, Xieng Khouang Province, Lao PDR. The primary purpose of the project is to generate electricity to the Lao Power Grid.

Total installed capacity of the project is 10.50 MW, provided by two (2) Horizontal Francis turbines. The scheduled commissioning date of the project activity is expected to be on 08-August-2024 with an average annual output of 44,270 MWh.

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e., electricity supplied from the power grid. By displacing part of the power generated by thermal power plants, the project is therefore expected to reduce an average annual emission reduction of 20,550 $tCO_2e$ per year with a total emission reduction of 143,851 $tCO_2e$ during the first crediting period.

The project is not only supplying renewable electricity to grid, but also producing positive environmental and economic benefits and also, promotes the local sustainable development in the following aspects:

- During the construction period, plenty of job opportunities were provided to local residents.
- The infrastructure has been greatly improved. The enhancement of the transportation and electricity system brings substantial benefits to local villagers.
- The use of fossil fuel will be reduced because of displacement by grid electricity, this reduces the fossil fuel-based electricity generation.
- Provide clean & cheap electricity in this region, promote the sustainable development in this region and slowing down the increasing trend of GHG emissions.

## 1.2 Audit History

Not Applicable, as the project is currently in the listing stage.

## 1.3 Sectoral Scope and Project Type

| | |
|---|---|
| Sectoral scope[1] | 01-Energy Industries (Renewable/non-renewable sources) |
| Project activity type | Renewable Energy Project |

This project is not an AFOLU project and is not a grouped project.

| | |
|---|---|
| Sectoral scope | NA |
| AFOLU project category[2] | NA |
| Project activity type | NA |

## 1.4 Project Eligibility

### 1.4.1 General eligibility

The project is a 10.50 MW small scale grid connected electricity generation project using hydro power in Lao PDR. which is least developed country (LDC)[3] as per the United Nation (UN). As per the VSC Standard version 4.6[4], section 2.1.3 – Table 1, small scale grid connected hydro power plants located in LDCs are within the VCS program scope, making this project eligible under VCS.

The project activity is yet to be commissioned, with the scheduled commissioning date is expected to be on 08-August-2024, Therefore, the project meets the requirements related to the pipeline listing deadline, the opening meeting with the validation body, and the validation deadline

The methodology (AMS.I. D version 18) applied for the project activity is eligible under the VSC program and the project is not fragmented part of a large project but rather an individual project implemented by the project owner.

### 1.4.2 AFOLU project eligibility

Not Applicable. This project is a hydro power project which falls under the non-AFOLU project category.

### 1.4.3 Transfer project eligibility

The project is yet to be commissioned and it is not registered under any other GHG programs.

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes:* https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

[2] *See Appendix 1 of the* VCS Standard

[3] https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf

[4] https://verra.org/wp-content/uploads/2024/03/VCS-Standard-v4.6.pdf

## 1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

### 1.5.1 Grouped project design

The project activity is not a grouped project. Thus, this section is not applicable.

## 1.6 Project Proponent

| | |
|---|---|
| **Organization name** | Nonghai Group Sole Company Limited |
| **Contact person** | Mr. Champa Khammanivong |
| **Title** | General Manager |
| **Address** | Kamphengmuang road, Nonghai Village, Hatsayphong Distric, Vientiane Capital, Lao PDR. |
| **Telephone** | +856 21 351 025 |
| **Email** | nonghaigroup@yahoo.com |

## 1.7 Other Entities Involved in the Project

| | |
|---|---|
| **Organization name** | Kosher Climate India Private Limited |
| **Role in the project** | Project Representative |
| **Contact person** | Vamsi Krishna M |
| **Title** | Director |
| **Address** | Zee Plaza, No.1678, 27th Main Rd, Near KLM Mall, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India |
| **Telephone** | +91 99453 43475 |
| **Email** | vamsi@kosherclimate.com<br>narendra@kosherclimate.com |

## 1.8 Ownership

The project owner is Nonghai Group Sole Company Limited. The business licence of the project owner are the evidences for legislative right. Besides, the project allotment license, the equipment purchasing contract and the power purchase agreement are the evidence for the ownership of the plant equipment and power generating.

## 1.9 Project Start Date

| | |
|---|---|
| **Project start date** | 08-August-2024 |
| **Justification** | As per VCS Standard v4.6, section 3.8, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions, which is the commissioning date of the project activity i.e., 08-August-2024. |

## 1.10 Project Crediting Period

| | |
|---|---|
| **Crediting period** | ☒ *Seven years, twice renewable*<br>☐ *Ten years, fixed*<br>☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)* |
| **Start and end date of first or fixed crediting period** | 08-August-2024 to 07-August-2031 |

## 1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 $tCO_2e$/year (project)

☐ ≥ 300,000 $tCO_2e$/year (large project)

| Calendar year of crediting period | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 08-August-2024 to 07-August-2025 | 20,550 |
| 08-August-2025 to 07-August-2026 | 20,550 |
| 08-August-2026 to 07-August-2027 | 20,550 |

| | |
|---|---|
| 08-August-2027 to 07-August-2028 | 20,550 |
| 08-August-2028 to 07-August-2029 | 20,550 |
| 08-August-2029 to 07-August-2030 | 20,550 |
| 08-August-2030 to 07-August-2031 | 20,550 |
| **Total estimated ERRs during the first or fixed crediting period** | **143,851** |
| **Total number of years** | **7** |
| **Average annual ERRs** | **20,550** |

## 1.12 Description of the Project Activity

The project activity is a run-of-river Hydro power project with a total installed capacity of 10.50 MW provided by two Horizontal Francis turbines. The project is promoted by Nonghai Group Sole Company Limited.

The Project activity is a new facility (Greenfield) and the electricity generated by the project will be exported to the electricity grid. The project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The Project Proponent plans to avail the VCS benefits for the project.

In the Pre- project scenario, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity.

The project shall result in replacing anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 20,550 $tCO_2e$ per year, thereon displacing 44,270 MWh/year amount of electricity from the gird over the 7 years crediting period.

The hydro project structure consists of release sluice, main powerhouse, and right-bank non overflow dam, etc.

**Table.1: Main Technical Parameters of the project**

| **Turbine** | | |
|---|---|---|
| **Item** | **Value** | **Unit** |
| The height of the highest-level waterfall | 74.35 | m |
| The height of the lowest- level waterfall | 66.50 | m |

| | | |
|---|---|---|
| Maximum flow rate | 18 | $m^3/s$ |
| Rated Head | 74.35 | m |
| Type | Horizontal Francis | - |
| Number of turbines | 2 | Nos |
| Rated Capacity | 10,500 | kW |
| Capacity per turbine | 5,250 | kW |
| Rated speed | 600 | r/m |
| **Generator** | | |
| **Item** | **Value** | **Unit** |
| Model | SF9-10/2150 | |
| Rated Capacity | 5,000 | kW |
| Number of generators | 2 | Nos |
| Power factor | 0.80 | |
| Rated Voltage | 6.3 | kV |

## 1.13 Project Location

The Nam Ngiep 2A Hydropower project is construction on Nam Ngiep River, Sanluang Town, Khoun District, Xieng Khouang Province, Lao PDR.

**Table.2: Project Location**

| Project Location | Latitude | Longitude |
|---|---|---|
| Sanluang Town, Khoun District, Xieng Khouang Province, Lao PDR. | 19°09'12.69"N (19.1535) | 103°20'41.41"E (103.3448) |

## 1.14 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site. For more information refer to section 3.4 for the baseline scenario.

## 1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has received all the necessary approvals for the development and commissioning of the proposed 10.50 MW hydro power project from the respective nodal agencies and is in compliance to the local laws and regulations. The list of applicable laws & regulations and the project compliance against it are given below:

| Laws & Regulation | Project Compliance |
|---|---|
| Law on Electricity (amended version) No. 03/NA, dated 20/12/2011[5]; | Project received Approval from Ministry of Mine and Energy that confirms the project compliance with these Laws & Regulation. The project also signed PPA with Lao Electric State Enterprise, a state-owned enterprise which is also an approval for electricity connection with the grid. |
| Decree on the Approval and Promulgation of the Policy on Sustainable Hydropower Development of the Lao PDR No. 02/GoL., dated 12/01/2015[6] ; | |
| Law on Environmental Protection (EPL), No. 29/NA, dated 18/12/2012[7]; | Project received Environmental clearance from Water Resources and Environment Agency (WREA) based on the IEE report submitted that confirms the project compliance with these laws & regulation. |
| National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR[8]. | |

## 1.16 Double Counting and Participation under Other GHG Programs

### 1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

5 https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf

6 policy.asiapacificenergy.org/sites/default/files/Decree%20on%20the%20Approval%20and%20Promulgation%20of%20the%20Policy%20on%20Sustainable%20Hydropower%20Development%20in%20Lao%20PDR.pdf

7 https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

8 policy.asiapacificenergy.org/sites/default/files/National%20Policy%20on%20the%20Environmental%20and%20Social%20Sustainability%20of%20the%20Hydropower%20Sector.pdf

### 1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

### 1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

## 1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

### 1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

### 1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

### 1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes ☒ No

## 1.18 Sustainable Development Contributions

The solar power project at hand plays a direct role in advancing Sustainable Development Goal (SDG) 7, specifically focusing on ensuring universal access to affordable, reliable, sustainable, and modern energy. This access to energy is pivotal for fostering sustainable economic growth and development, addressing climate action, promoting energy transformation, and enhancing equity and inclusiveness in a cohesive manner.

**Table 3: Contributions of the project activity to SDGs**

| | |
|---|---|
|  | By generating an estimated annual output of 20,550 MWh of renewable energy, the project significantly contributes to the realization of SDG 7, which strives to ensure access to affordable, reliable, sustainable, and modern energy for all. This contribution aligns with the overarching goal of fostering a greener and more sustainable future. |
|  | Throughout the construction and operational phases, the project will create numerous direct and indirect employment opportunities, predominantly benefitting the local population in the region. This proactive approach contributes to the fulfilment of SDG 8, which seeks to promote sustained, inclusive, and sustainable economic growth, along with full and productive employment and decent work for all. |
|  | Through its efforts in emission reductions, the project aligns with the objectives of SDG 13, which calls for urgent action to combat climate change and its adverse impacts. By taking substantial measures to decrease greenhouse gas emissions, the project actively contributes to the global initiative of safeguarding the planet for the well-being of future generations. |

## 1.19 Additional Information Relevant to the Project

### 1.19.1 Leakage Management

Not applicable as no leakage emission is involved in this project activity.

### 1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

### 1.19.3 Further Information

Not applicable, as this project activity does not involve any additional information that may have a bearing on the eligibility of the project, the GHG emission reductions, or the quantification of the project's reductions.

# 2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

## 2.1 Stakeholder Engagement and Consultation

### 2.1.1 Stakeholder Identification

| | |
|---|---|
| *Stakeholder Identification* | Stakeholders include various individuals, groups, and institutions who have an interest in, could be impacted by the project, possess the ability to significantly influence, or are crucial for achieving the desired project outcome. The list of relevant stakeholders identified through stakeholder mapping includes:<br>• Local people, communities, and representatives expected to be directly or indirectly affected by or interested in the project.<br>• Local policymakers and representatives of local authorities.<br>• Local non-governmental organizations (NGOs) and Women Groups focusing on topics relevant to the project or collaborating with communities likely to be affected by the project.<br>Stakeholders have been invited through various means of invitation like public announcement, invitation letters, etc. |
| Legal or customary tenure/access rights | The project is built own and legally owned by Nonghai Group Sole Company Limited, and the project has been constructed on the own land belonging to Nonghai Group Sole Company Limited. Nonghai Group Sole Company Limited has 100% legal and customary rights on the project activity over and above the project lifetime. Construction of project activity does not lead to rehabilitation of any indigenous people or local communities.<br>There is no forceful settlement or rehabilitation that occurred due to the project implementation. No other stakeholders or indigenous |

| | |
|---|---|
| | people has the legal right on the project area prior or post construction of project activity. |
| Stakeholder diversity and changes over time | The Nam Ngiep 2A Hydropower project is located in Sanluang Town, Khoun District, Xieng Khouang Province. A socio-economic survey was carried out at Khoun District, Xieng Khouang Province to determine relevant demographic characteristics of the communities. Based on the socio-economic profile, stakeholder mapping was carried out to ensure that equitable, non-discriminatory, and meaningful consultation was carried out.<br><br>The stakeholder group exhibited notable diversity across social, economic and cultural dimensions. Socially, individuals from various backgrounds, including different age groups, ethnicities and socio-economic classes were present. Overtime, the composition of these groups may undergo changes due to evolving circumstances and development.<br><br>However, project activity does not lead to any stakeholder's diversity as the hydro power project does not have any direct or indirect impact on the local stakeholders culture or diversity.<br><br>However, to ensure that the project continues to align with the diverse needs and expectations of its stakeholders, the project proponents has implemented a Grievance Redressal Mechanism (GRM). |
| Expected changes in well-being | As a result of the hydro power plant some of the closest proximity communities may lose some of their income generation activities such as fishing, income from river weed, due to the hydro power plant. To address the same project owner will compensate those communities with resettlement and rehabilitation compensation at the time of project start date and at possible extent provide the income generation activities during the construction and operation of the project activity. |
| Location of stakeholders | The location of the stakeholders is – Sanluang Town, Khoun District, Xieng Khouang Province.<br><br>There are no areas predicted to be impacted by the project. |
| Location of resources | Other than project owners and working employees no other general stakeholders or local communities have access to the project site. |

| | |
|---|---|
| | Though local communities have the access to the river and water ponds for the fishing activities those are not attained by legal or through customary rights. |

### 2.1.2 Stakeholder Consultation and Ongoing Communication

| | |
|---|---|
| Date of stakeholder consultation | 02- March-2019 |
| Stakeholder engagement process | Key stakeholders were identified by stakeholder mapping considering social, cultural and gender-specific factors that might influence their engagement. Representation and participation from diverse groups such as local communities, local governments, national & local NGOs, private sector entities comprising women, men, elderly, girls and boys etc. was sought by inviting them to attend the meeting through different means such as: |
| Consultation outcome | Initial consultations with relevant stakeholders potentially affected by the project were conducted in march 2019 by Nonghai Group Sole Company Limited to inform them of the proposed project, potential temporary impacts and project benefits.<br><br>The objective of the meeting was based on the non-technical summary, Environmental Management Plan and draft Passport Report of the project. Also, how the project might have some environmental effects, how these issues will be mitigated by the investor and also climate change and how the project will help the fight against climate change were discussed.<br><br>All of participants supported the project and all the questions raised by the stakeholders were clarified by the project owner. No concerns raised by the stakeholders that require change in project design |
| Ongoing communication | There is a Grievance Redress Mechanism (GRM) to receive and facilitate resolution of project affected persons' concerns, complaints, and grievances about the project's environmental and social performance. |
| Stakeholder input | During the consultation, the project proponents discussed potential impacts, and gave stakeholders the chance to express concerns and suggestions for improvement. |

| | |
|---|---|
| | Stakeholders expressed satisfaction with the project outcomes and comments raised were mainly on the job opportunities due to the project activity among the local people. The project proponents assured local community first preference for the employment opportunities during construction and operation of the project. Concerns about the effect of project on the water quantity and quality were addressed.<br><br>Stakeholder comments mainly sought clarification, and as sufficient information was provided during the meeting, no changes to the project design are deemed necessary. |

### 2.1.3 Free Prior and Informed Consent

| | |
|---|---|
| **Obtaining consent** | The project activity has been implemented with all necessary approvals from Electricite Du Laos (EDL). Project proponent has conducted LSC during which all the necessary information regarding the project activity was explained to the stakeholders.<br><br>The project proponents fostered open dialogues, providing stakeholders ample opportunities to express concerns, ask questions, and share input. All raised concerns and inputs have been documented, and a grievance resolution mechanism is in place to address any ongoing conflicts, demonstrating the project's commitment to prioritizing resolution and community harmony. Regular stakeholder engagement will persist throughout the project lifecycle to address evolving concerns and ensure ongoing support.<br><br>Through this comprehensive process, the project not only obtained consent but also built a foundation of trust and cooperation with stakeholders, resulting in no ongoing or unresolved conflicts. |
| **Outcome of FPIC** | The result of the FPIC and stakeholder consultation is positive, with stakeholders expressing their support for the project. Project proponent has provided resettlement compensation to the affected if any, ensuring that the project did not encroach on land, relocate people without consent, or cause forced physical or economic displacement |

### 2.1.4 Grievance Redress Procedure

| | |
|---|---|
| **Development process** | Grievance Redress Mechanism (GRM) has been established to address concerns raised by stakeholders. This involves a Grievance |

| | |
|---|---|
| | Redress Committee (GRC), consisting of relevant commune and/or village chiefs, provincial officers or their nominees, a District Officer from the Cadastral office or their nominee, the Contractor, and a witness of the affected person, with at least one female member.<br><br>The GRC is tasked with (i) resolving land acquisition issues (if any), (ii) addressing concerns related to dust, noise, vibration, and construction-related nuisances to the public, (iii) convening monthly to review lodged complaints (if any), (iv) recording grievances and resolving them within 30 days from filing, and (v) reporting the status of grievance resolution and decisions/actions taken to the complainant(s)/affected persons.<br><br>The Provincial GRC, established at the provincial level, includes representation from commune councils across the project area of influence where project components will be implemented. Key contacts for the GRC have been prominently displayed at project office sites and public notice boards in affected communes, in Khmer language, to ensure accessibility and awareness. |
| **Grievance redress procedure** | Complaints will be recorded in the Grievance Record and it will be further considered by the project proponent to discuss and resolve the issue raised by the stakeholders.<br><br>Recordkeeping and Reporting: Project proponent will maintain records of grievances, including complainant details, complaint date, nature of grievance, corrective actions, and resolution dates. |

### 2.1.5 Public Comments

This section will be updated after the commenting period is completed.

## 2.2 Risks to Stakeholders and the Environment

| | Risks identified | Mitigation or preventative measure taken |
|---|---|---|
| **Risks to stakeholder participation** | No risk identified | No risks to stakeholder participation were identified during the preparation for stakeholder consultation. The project developer analyzed relevant stakeholders, risks, opportunities, and strategies for |

| | | |
|---|---|---|
| | | conducting the most effective consultation. To ensure inclusivity, the consultation venues were chosen centrally and made accessible to people of all castes, creeds, and sects in the society, ensuring that the location was considered safe by everyone in the community. Focus Group Discussions were conducted specifically to provide women and marginalized groups with an opportunity to express their opinions. |
| **Working conditions** | No risk identified | Occupational Health and Safety will be prioritized at every stage of the project. Employees will receive regular training to ensure their health and safety in the workplace. Measures will also be implemented to safeguard the health of the community near the project area. The project is committed to complying with national Occupational Health and Safety Standards[9], national labour standards[10], IFC PS2[11] and International Labour Organization Labor Standards, following all guidelines and standards. |
| **Safety of women and girls** | No risk identified | The project promotes safe and secure environments for all stakeholders including women and girls. |

[9] https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=73666

[10] https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=96369

[11] https://www.ifc.org/content/dam/ifc/doc/2010/2012-ifc-performance-standard-2-en.pdf

| | | |
|---|---|---|
| | | The proposed project is a renewable energy grid connected power project and therefore it's not gender sensitive and does not adversely impact the safety of women and girls |
| Safety of minority and marginalized groups, including children | No risk identified | The project promotes safe and secure environments for all stakeholders including minority, marginalized groups and children |
| Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials) | No risk identified | The project is a hydro power project, hence will not result in release of pollutants to the environment.<br>The project will have a temporary impact on the environment only during the construction phase due to soil disruption, particulate dust generation and noise associated with vehicles, respective measures have been taken to minimize the impact. |

## 2.3 Respect for Human Rights and Equity

### 2.3.1 Labor and Work

| | |
|---|---|
| Discrimination and sexual harassment | The project is committed to ensuring a work environment free from discrimination and sexual harassment. The project developer has a non-harassment policy in place to prohibit any form of discrimination or harassment based on factors such as race, gender, age, religion, or any other protected characteristic. All employees are made aware of these policies through training programs, and clear reporting mechanisms are in place to address any concerns or incidents promptly. |
| Management experience | Nonghai Group Sole Company Limited is implementing the project and the company is having experience in implementing similar project activities and engaging communities. |
| Gender equity in labor and work | The project developer has a Non-Discrimination and Equal Opportunity Policy in place, promoting equal opportunities, non-discrimination, and |

| | |
|---|---|
| | fair treatment of workers. The project is committed to eliminating gender-based discrimination and ensuring that everyone, regardless of gender, has equal access to employment, training, and advancement opportunities. Moreover, the project follows transparent and non-discriminatory hiring practices and has set up a fair compensation structure based on principles of pay equity, ensuring that individuals receive equal pay for equal work. |
| **Human trafficking, forced labor, and child labor** | Nonghai Group Sole Company Limited, adhere to all national and international laws related to child labor, forced labor, and human trafficking. The project unequivocally affirms that it does not and will not use victims of human trafficking, forced labor, or child labor in any aspect of its operations. This commitment aligns with the project's dedication to upholding ethical and legal standards in its workforce practices |

### 2.3.2 Human Rights

The project avoids any activities that could violate the economic, social, and cultural well-being of local communities. Consequently, the project developer and the project itself commit to respecting internationally proclaimed human rights. They will not be complicit in any form of violence or human rights abuses, as defined in the Universal Declaration of Human Rights. This commitment reflects the project's dedication to upholding ethical standards and safeguarding the well-being of the communities involved.

### 2.3.3 Indigenous Peoples and Cultural Heritage

The Project will not directly or indirectly affect the dignity, human rights, livelihood systems, or culture of Indigenous Peoples (IPs), neither will it affect the territories or natural or cultural resources that IPs own, use, occupy, or claim as their ancestral domain.

### 2.3.4 Property Rights

| | |
|---|---|
| **Rights to territories and resources** | Project owner is owned the rights on the project equipment, land, transmission and distribution network.<br><br>Other rights on river water and surrounding natural resources were vested with the government or local communities. Project activity will not change the natural rights on territories and natural resources and has no impact over the future. |
| **Respect for property rights** | Not applicable as the project does not involve any territories or resources to which the stakeholders have customary rights or access. |

### 2.3.5 Benefit Sharing

As mentioned earlier the project is owned by the Nonghai Group Sole Company Limited and the property rights vest within their control therefore benefit sharing is not applicable and hence no benefit sharing plan has been designed.

| | |
|---|---|
| Process used to design the benefit sharing plan | NA |
| Summary of the benefit sharing plan | NA |
| Approval and dissemination of benefit sharing plan | NA |

## 2.4 Ecosystem Health

| | Risks identified | Mitigation or preventative measure taken |
|---|---|---|
| Impacts on biodiversity and ecosystems | No risk identified | The project is not located in sensitive ecological zones, biodiversity conservation areas, and there are no rare and valuable plant and animal |
| Soil degradation and soil erosion | Risk of soil erosion during construction | The project might lead to soil erosion during the construction phase. However, mitigation measures will be implemented, such as recovering topsoil stripped during clearing and construction. Land clearing will be confined to the specific project site needed for development, minimizing the impact. |
| Water consumption and stress | No risk identified | The water supply during construction and operational phase |

| | | |
|---|---|---|
| | | is taken from the surface water of reservoir. |
| Usage of fertilizers | No risk identified | The project harnessed hydro energy to generate electricity and therefore, will not involve the application of fertilizers, pesticides or herbicides |

### 2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

### 2.4.2 Introduction of species

This is not a AFOLU project and doesn't involve introduction of any species into the project area, hence this section is Not Applicable. Also, there are no threatened livelihood in the vicinity of the project activity

| Species introduced | Classification | Justification for use | Adverse effects and mitigation |
|---|---|---|---|
| NA | NA | NA | NA |

| Existing invasive species | Mitigation measures to prevent spread or continued existence of invasive species |
|---|---|
| NA | NA |

### 2.4.3 Ecosystem conversion

The project activity has not resulted in conversion or clearing of the existing ecosystem

# 3 APPLICATION OF METHODOLOGY

## 3.1 Title and Reference of Methodology

| Type (methodology, tool or module). | Reference ID, if applicable | Title | Version |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Methodology | AMS.I. D[12] | Small-scale methodology for grid-connected renewable electricity generation. | 18.0 |
| Tool | 07[13] | Tool to calculate the emission factor for an electricity system. | 07.0 |
| Tool | 21[14] | Demonstration of additionality of small-scale project activities | 13.1 |
| Tool | 27[15] | Investment Analysis | 13.0 |

## 3.2 Applicability of Methodology

The project activity involves generation of grid connected electricity from hydro energy. The project activity has a proposed capacity of 10.50 MW which will qualify for a small scale CDM project activity under Type-I of the small-scale methodologies. The project status is corresponding to the methodology AMS I.D version 18.0 and applicability of methodology is discussed below.

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| AMS.I. D | This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass:<br><br>(a) Supplying electricity to a national or a regional grid; or<br><br>(b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling. | The project activity involves setting up of a renewable energy (hydro) generation plant that exports electricity to the fossil fuel dominated electricity grid. Thus, the project meets this applicability conditions "a" |
| | Illustration of respective situations under which each of the methodology (i.e., "AMS-I.D.: Grid connected renewable electricity generation", "AMS-I.F.: | According to the point 1 of the Table 2 in the methodology – "Project supplies electricity to a national/ regional grid" is |

[12] https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI9ODBC

[13] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[14] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf

[15] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf

| | | |
|---|---|---|
| | Renewable electricity generation for captive use and mini-grid" and "AMS-I.A.: Electricity generation by the user) applies is included in the appendix. | applicable under AMS I.D. As the project activity supplies the electricityto Lao PDR Grid system grid which is a regionalgrid, the methodology AMS-I.D. is applicable. |
| | This methodology is applicable to project activities that:<br><br>(a) Install a Greenfield plant;<br><br>(b) Involve a capacity addition in (an) existing plant(s);<br><br>(c) Involve a retrofit of (an) existing plant(s);<br><br>(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or<br><br>(e) Involve a replacement of (an) existing plant(s). | The Project activity involves the installation of new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity. Thus, Project activity is a Greenfield plant and satisfies this applicability condition (a). |
| | Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:<br><br>(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;<br><br>(b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 $W/m^2$;<br><br>(c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 $W/m^2$ | As the project activity is a run-off river type hydro power plant, this criterion is not relevant for the project activity. |

| | | |
|---|---|---|
| | If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW | The rated capacity of the project activity is 10.50 MW with no provision of Co-firing fossil fuel. Hence, meeting with this criterion. |
| | Combined heat and power (co-generation) systems are not eligible under this category | This is not relevant to the project activity as the project involves only hydro power generating units. |
| | In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct1 from the existing units. | There is no other existing renewable energy power generation facility at the project site. Therefore, this criterion is not applicable. |
| | In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW. | The project activity is a new installation, it does not involve any retrofit measures nor any replacement and hence is not applicable for the project activity. |
| | In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered | This project is a hydro power project activity and hence this criterion is not applicable. |

| | | |
|---|---|---|
| | methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored. | |
| | In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply. | This project is a hydro power project activity and hence this criterion is not applicable. |
| **In addition, the above applicability conditions, the applicability conditions of tool referred in the methodology AMS I.D, version 18.0 has been referred here under:** | | |
| Tool 07 | This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes of grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g.: demand-side energy efficiency projects). | The project activity is a Greenfield hydro power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system". |
| | Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off- grid power plants. In the latter case, the conditions specified in "Appendix 2: Procedures related to off- grid power generation" should be met. Namely, the total capacity of off-grid power plants (in | Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly. |

| | | |
|---|---|---|
| | MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity. | |
| | In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. | The project activity is located in Lao PDR, a non- Annex I country. Therefore, this criterion is not applicable for the project activity |
| | Under this tool, the value applied to the $CO_2$ emission factor of bio fuels is zero. | The project activity is a grid connected hydro power project and therefore, this criterion is not applicable for the project activity |
| Tool 21 | The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board | Since the applied technology is not a new methodology project proponent has applied this tool for the demonstration additionality in compliance with the tool. Refer to section 3.5, for the detailed applicability of this tool and additionality assessment.<br><br>Hence this tool is applicable. |
| Tool 27 | This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool | PP used "demonstration of additionality of small-scale project activities" (formerly named as "non-binding best practice examples to |

| | | |
|---|---|---|
| | "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. | demonstrate additionality for SSC project activities" to demonstrate additionality of the project. Hence, using Tool 27 for conducting investment analysis is appropriate. |
| | In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. | The applied methodology AMS I.D does not contain requirement for the investment analysis. Hence this criterion is not applicable |

## 3.3 Project Boundary

As per applicable methodology AMS-I.D. Version 18, "The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the power plant is connected to."

Thus, the project boundary includes the Hydro Turbine Generators and the Lao grid system.

The project does not involve any other emission sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

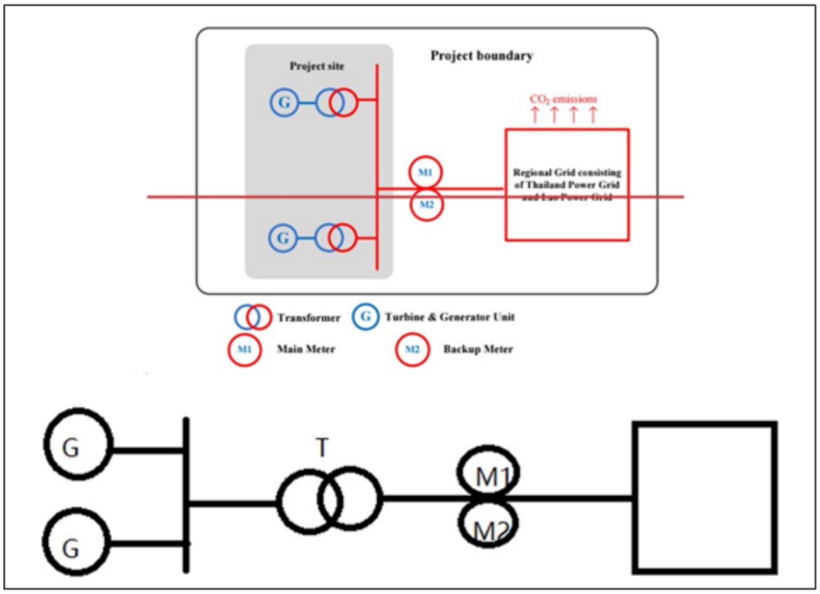


Figure 2: Project Boundary

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| Baseline | Source 1 | $CO_2$ | Yes | Main Emission Source |
| | | $CH_4$ | No | Minor Emission Source |
| | | $N_2O$ | No | Minor Emission Source |
| | | Other | No | NA |
| Project | Source 1 | $CO_2$ | No | Project activity does not emit $CO_2$ |
| | | $CH_4$ | No | Project activity does not emit $CH_4$ |
| | | $N_2O$ | No | Project activity does not emit $N_2O$ |
| | | Other | No | NA |

## 3.4 Baseline Scenario

As per the approved consolidated Methodology AMS I.D (Version 18.0) para 19: "If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The project activity involves setting up of hydro project to harness the power of hydro energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the national grid, which is fed mainly by fossil fuel fired plants.

The project is run-of-river hydropower project. the project electricity system is a regional grid consisting of Thailand Power Grid and the Lao Power Grid. In the absence of the project, the local was/will be supplied by the above-mentioned regional grid. Thus, the baseline scenario of the project is continuation of the present situation, i.e., electricity supplied from the regional power grid.

The combined margin of the grid used for the project activity is as follows:

| Parameter | Value | Nomenclature |
|---|---|---|
| $EF_{grid,CM,y}$ | 0.5542 $tCO_2/MWh$ | Combined margin $CO_2$ emission factor for the project electricitysystem in year y. |
| $EF_{grid,OM,y}$ | 0.5542 $tCO_2/MWh$ | Operating margin $CO_2$ emission factor for the project electricitysystem in year y. |
| $EF_{grid,BM,y}$ | 0 $tCO_2/MWh$ | Build margin $CO_2$ emission factor for the project electricity systemin year y. |

## 3.5 Additionality

### 3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☒ Yes ☐ No

### 3.5.2 Additionality Methods

The additionality of a VCS Project shall be demonstrated by applying the following approach having two components:

(i) A Legal Requirement Test; and

(ii) An Additionality Test.

The project activity does not fulfil the criteria of positive list as provided in CDM Tool 32: "Methodological Tool – Positive List of Technologies" and hence additionality of the project activity is demonstrated through a project specific additionality test.

For the demonstration and assessment of additionality, Tool 21 "demonstration of additionality of small- scale project activities", Version 13.1 has been applied. As per the para 10 of the tool, project owner shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

**Investment Barrier:** a financially more viable alternative to the project activity would have led to higher emissions

**Technological barrier:** a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity as so would have led to higher emissions:

Barrier due to prevailing practice: or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions.

**Other barriers:** without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

The project investor has selected Investment barrier to demonstrate in a conservative and transparent manner that the proposed VCS project activity is financially unattractive.

To conduct the investment analysis, Methodological tool 27: Investment analysis, version 13, EB 120 Annex 3 has been referred.

Considering the fact that the alternative to the project is the supply of electricity from the grid & the choice of the developer is to invest or not to invest, benchmark analysis has been considered appropriate for demonstration of additionality, which is in conformity with "Investment Analysis" Annex 3 EB 120.

**Apply benchmark analysis**

As per para 15 of Tool 27: Investment analysis, Version 13 states that Required/expected returns on equity are appropriate benchmarks for equity IRR. The project participant has chosen benchmark analysis to demonstrate the additionality of the project. The project is promoted by private limited company and hence the return on equity and the risks associated with the investments for their shareholder is of primary concern. Hence, in order to analyse the financial viability of the project activity, the prime financial indicator that has been used is the post-tax equity IRR of the project activity.

**Selection of Appropriate Benchmark**

The benchmark has been considered in accordance with Guidance 19 of EB 120 Annex 3, "The values in the table in the Appendix may also be used, as a simple default option".

Methodology deployed for arriving at a suitable value of Benchmark using Default Value has been described below:

- As the proposed project activity generates power utilizing hydro energy, Group 1 as per para 5a of Appendix of EB 120 Annex 3 has been identified as a suitable category.
- The investment analysis has been carried out in Nominal terms. Accordingly, default value as given in table under the Appendix, Tool 27 has been adjusted by adding suitable forecasted inflation rate taken from IMF world Economic database as no inflation forecast or inflation target published by Lao PDR.
- Average forecasted inflation rate for the host country (Lao PDR) published by the IMF (International Monetary Fund) World Economic Outlook for the next five years has been considered.

The decision-taken for the project is on 15-August-2015. Hence applicable inflation rate has been chosen from the IMF databases accordingly for the estimation of resultant benchmark.

The benchmark has been computed in the following manner:

**Default Value Benchmark:**

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e., Default values for cost of equity (expected return on equity) in the 'Methodological tool: Investment analysis.

The Required return on equity (benchmark) was computed in the following manner:

**Nominal Benchmark7 = {(1+Real Benchmark) * (1+Inflation rate)}-1**

Where:

Default value for Real Benchmark = 20.98% (CDM Tool 27: Investment Analysis, version 13)

Inflation Rate forecast for by (International Monetary Fund) World Economic Outlook database for Lao PDR.

**Benchmark estimation:**

The Cost of Equity has been considered using the "Methodological tool: Investment analysis" available at the time of decision making as well as the latest available value. As a conservative approach, the minimum value of benchmark has been considered as calculated using these 2 approaches.

Table under Appendix in EB120, Annex 3 specifies default value of expected return on equity in real terms for Energy Industries in Lao PDR = 20.98%[16]

Thus, minimum cost of equity considered for calculation of Benchmark = 20.98%

5-year inflation Forecast for Lao PDR as IMF World Economic Outlook available at the time of investment decision and corresponding benchmark values is:

| Inflation forecast for 5 years | Source |
|---|---|
| 2.59% | World Economic Outlook database: Apr 2016[17] |

Corresponding benchmark values applicable at the time of investment decision time: Nominal Benchmark = {(1+20.98%) * (1+2.59%)}-1 = 24.11%

Calculation and comparison of financial indicators.

The period considered for Post Tax Equity IRR calculations is 30 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Varying and fixed Input values considered for the IRR calculation are provided below.

| Particulars | Value | Unit | Source/Remarks |
|---|---|---|---|
| Capacity of the project | 10.5 | MW | Basic Design Report |
| PLF | 48.1% | % | Calculated |
| Annual Net generation | 44.27 | GWh | Basic Design Report |
| Project cost | 16.93 | Mn USD | Basic Design Report |
| Debt | 80% | % | Basic Design Report |
| Equity | 20% | % | Basic Design Report |
| Debt | 13.54 | Mn USD | Calculated |

[16] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf

[17] https://www.imf.org/en/Publications/WEO/weo-database/2016/April/weo-report?c=544,&s=PCPIPCH,PCPIEPCH,&sy=2017&ey=2021&ssm=0&scsm=1&scc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1

| | | | |
|---|---|---|---|
| Equity | 3.39 | Mn USD | Calculated |
| Interest rate | 7.43% | % | Basic Design Report |
| Debt Repayment tenure | 10 | years | Basic Design Report |
| Moratorium | 0 | year | Basic Design Report |
| O & M cost | 3.00% | % | Basic Design Report |
| O & M cost | 0.51 | Mn USD | Calculated |
| Insurance Cost | 0.04 | Mn USD | Basic Design Report |
| Labour Cost | 0.01 | Mn USD | Basic Design Report |
| tariff | 0.065 | USD/kWh | Basic Design Report |
| Depreciation Rate | 4.00% | % | Basic Design Report |
| Salvage Value | 10% | % | Basic Design Report |
| Income tax rate | 15% | % | Basic Design Report |
| Salvage value in USD million | 1.69 | Mn USD | Calculated |
| Residual value | 1.69 | Mn USD | Calculated |

Post Tax Equity IRR for the project activity against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is below the benchmark value.

| | |
|---|---|
| Post tax Equity IRR | 15.29% |
| Benchmark Value | 24.11% |

**Sensitivity Analysis**

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 3 of EB 120, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. PP has identified the total revenue from the project activity is dependent on the Tariff, Annual Net Generation, Project Cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

| Parameter | -10% | Normal | 10% | Variation required to reach benchmark | Value required to reach benchmark |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Tariff (USD/kWh) | 12.07% | 15.29% | 18.89% | 22.70% | 0.080 |
| Net Generation (GWh) | 12.07% | 15.29% | 18.89% | 22.70% | 54.319 |
| Project Cost (Mn USD) | 18.55% | 15.29% | 12.87% | -21.70% | 13.25 |
| O&M cost (Mn USD) | 15.90% | 15.29% | 14.70% | NA | NA |

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the proposed project activity is additional with reasonable variation in values and is not likely to reach the benchmark value.

**Conclusion**

Even with the changes of +/- 10% of the variation, the IRR is still less than the benchmark. Hence, the project is additional.

## 3.6 Methodology Deviations

There is no methodology deviation.

# 4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

## 4.1 Baseline Emissions

Baseline emissions include only $CO_2$ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$\mathbf{BE_y = EG_{PJ,y} \times EF_{grid,y}}$ (Equation 1)

Where:

$BE_Y$= Baseline emissions in year y (t $CO_2$)

$EG_{PJ,Y}$= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin $CO_2$ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t $CO_2$/MWh)

If the project activity is the installation of a greenfield power plant, then:

$EG_{PJ,Y} = EG_{PJ,facility,y}$ (Equation 2)

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh). The expected annual net electricity generation as per the FSR is 44,270 MWh

Calculate the $EF_{grid,y}$

The emission factor should be calculated in a transparent and conservative manner according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system” (Version 07.0).

The data used for calculation are from an official source (where available) and publicly available. The calculation processes are as follows:

STEP1: Identify the relevant electricity system;

STEP2: Choose whether to include off-grid power plants in the project electricity system (optional);

STEP3: Select a method to determine the operating margin (OM);

STEP4: Calculate the operating margin emission factor according to the selected method;

STEP5: Calculate the build margin (BM) emission factor;

STEP6: Calculate the combined margin (CM) emissions factor.

**STEP 1: Identify the relevant electricity system**

There are no transmission constraints between Lao and Thailand, the project electricity system is a regional grid consisting of Thailand Power Grid and Lao Power Grid. And as electricity imported from Malaysia, China and Vietnam Power Grid, these three Power Grids are considered as connected electricity system.

**STEP 2: Choose whether to include off-grid power plants in the project electricity system (optional)**

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I is chosen for emission factor calculation.

**STEP 3: Select a method to determine the operating margin (OM)**

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are four methods for calculating the $EF_{grid,OM,y}$:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch data Analysis OM, or

(d) Average OM

The method (d), average OM, is selected.

$EF_{grid,OM,ave,y}$ is calculated using ex ante option (d) 3-year generation-weighted average in 2020, 2019, 2018 without requirement to monitor and recalculate the emissions factor during the crediting period.

**STEP 4: Calculate the operating margin emission factor according to the selected method**

The average OM emission factor is calculated as the average emission rate of all power plants serving the grid, using the methodological guidance as described under Step 4 in the “Tool to calculate the emission factor for an electricity system” (Version 07.0) for the simple OM, but also including the low-cost / must-run power plants in all equations.

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options based on different data for calculating average OM:

Option A: Based on the net electricity generation and a $CO_2$ emission factor of each power unit; Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

For the project, the necessary data for Option A is not available, so Option B was used.

Under this option, the average OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$EF_{grid,OM\text{-}ave,y}$ = Average operating margin $CO_2$ emission factor in year y ($tCO_2$ /MWh)

$FC_{i,y}$ = Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ / mass or volume unit) $EF_{CO2,i,y}$ = $CO_2$ emission factor of fossil fuel type i in year y ($tCO_2$/GJ)

$EG_y$ = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)

I = All fossil fuel types combusted in power sources in project electricity system in year y = The data available in the most recent 3 years

According to the “Tool to calculate the emission factor for an electricity system” (Version 07.0), electricity imports from the connected electricity systems $EG_{import,y}$ are included in the $EG_y$.

The calculation value of average OM emission factor is 0.5542 $tCO_2$/MWh and the detailed calculating procedures please refer to ER calculation sheet.

**Step 5. Calculate the build margin (BM) emission factor**

To calculate the build margin (BM) emission factor, the data for determine the sample group of power units m about the most recently units in the electricity system is needed. However, as an international project system, it’s difficult to obtain the information for all the units in both Lao and Thailand (power generation data, commissioning date, and the fuel consumption). The data requirements for the application for calculate the build margin (BM) emission factor cannot be met.

As the Simplified CM is adopted in the step 6, the weighting of build margin emissions factor is 0.

**STEP 6: Calculate the combined margin (CM) emissions factor**

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

According to “Tool to calculate the emission factor for an electricity system” Version 07.0, the simplified CM can be used if:

*(a)* The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

(b) The data requirements for the application of Step 5 above cannot be met.

The project located in Lao, which is a Least Developed Country, therefore the criteria (a) is met; As mentioned in step 5, the data requirements for the application for calculate the build margin (BM) emission factor is not available, therefore the criteria (b) is also met.

The Simplified CM method is calculated as follow:

$$EF_{grid,CM,y}=w_{OM}\times EF_{grid,OM,y}+w_{BM}\times EF_{grid,BM,y}$$

Where:

$w_{OM}$ = Weighting of operating margin emission factor (%)

$w_{BM}$= Weighting of build margin emission factor (%).

The weighs $w_{OM}$ and $w_{BM}$, for simplified CM by default, are $w_{OM}$=1 and $w_{BM}$= 0. Hence calculation value of CM emission factor is 0.5542 $tCO_2$/MWh

Hence the baseline emission is calculated as below:

$BE_y$= 44,270× 0.5542 = 24,534 $tCO_2$

## 4.2 Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources” version 21[18] of paragraph no. 35,

$$PE_y = PE_{FF,y}+PE_{GP,y}+PE_{HP,y}+PE_{BESS,Y} \qquad \textbf{(Equation 1)}$$

The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

$PD$ = Power density of the project activity ($W/m^2$)

$Cap_{PJ}$ = Installed capacity of the hydro power plant after the implementation of the project activity (W) (i.e., 1,05,00,000 W)

$Cap_{BL}$ = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

[18] https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9

$A_{PJ}$ = Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full ($m^2$) (i.e., as per the satellite map it is 1,780,000 $m^2$)

$A_{BL}$ = Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full ($m^2$). For new reservoirs, this value is zero.

PD = (1,05,00,000 -0)/ (1,780,000 -0) = 5.9 W/$m^2$

Since the power density is greater than 4 W/$m^2$ but less than 10 W/$m^2$, the power density is calculated as below:

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$

Where,

$PE_{HP,y}$= Project emissions from water reservoirs (t $CO_{2e}$/yr)

$EF_{Res}$ = Default emission factor for emissions from reservoirs of hydro power plants (kg $CO_{2e}$/MWh) i.e., 90 kg $CO_{2e}$/MWh as per the methodology

$TEG_y$ = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh) 44,270 MWh/year as per project design

$PE_{HP,y}$= 90*44,270/1000 = 3,984 $tCO_2$

## 4.3 Leakage Emissions

No leakage emissions are considered.

## 4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The ex-ante emission reductions ($ER_y$) for the project activity are calculated as follows

$$\mathbf{ER_y = BE_y - PE_y - LE_y}$$

Where,

$ER_y$ = Emission Reduction in year y ($tCO_2$)

$BE_y$ = Baseline emission in year y ($tCO_2$)

$PE_y$ = Project emissions in year y ($tCO_2$)

$LE_y$ = Leakage Emissions in year y ($tCO_2$):

| Vintage period | Estimated baseline emissions ($tCO_2e$) | Estimated project emissions ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated reduction VCUs ($tCO_2e$) | Estimated removal VCUs ($tCO_2e$) | Estimated total VCUs ($tCO_2e$) |
|---|---|---|---|---|---|---|
| 08-August-2024 to 07-August-2025 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2025 to 07-August-2026 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2026 to 07-August-2027 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2027 to 07-August-2028 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2028 to 07-August-2029 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2029 to 07-August-2030 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| 08-August-2030 to 07-August-2031 | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |
| Total | 24,534 | 3,984 | 0 | 20,550 | 0 | 20,550 |

# 5 MONITORING

## 5.1 Data and Parameters Available at Validation

| | |
|---|---|
| **Data / Parameter** | $FC_{i,y}$ |
| **Data unit** | mass or volume unit of the fuel *i.* |

| | |
|---|---|
| Description | Amount of fossil fuel type i consumed in the project electricity system in year y. |
| Source of data | Fuel consumption in Lao PDR is calculated based on the Sustainability Report 2020, Banpu Power Public Company Limited[19].<br>Fuel consumption in Thailand is from “Energy Balance of Thailand 2020”, EGAT. |
| Value applied | See TABLE 1 in appendix 1. |
| Justification of choice of data or description of measurement methods and procedures applied | Data are from Banpu Power Public Company Limited (shareholder of Hongsa power plant) and Thailand authority,EGAT. |
| Purpose of data | Calculation of baseline emissions. |
| Comments | This parameter is fixed ex-ante for the entire crediting period. |

| | |
|---|---|
| Data / Parameter | $EF_{CO2,i,y}$ |
| Data unit | $tCO_2/TJ$ |
| Description | The $CO_2$ emission factor per unit of fuel i in year y |
| Source of data | 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 Chapter 1 Table 1.4[20] |
| Value applied | See the Table 2 in Appendix 1. |
| Justification of choice of data or description of measurement methods and procedures applied | No specific local value available, the value form IPCC 2006, Guidelines for National Greenhouse Gas Inventories was adopted. |
| Purpose of data | Calculation of baseline emissions. |
| Comments | This parameter is fixed ex-ante for the entire crediting period. |

| | |
|---|---|
| Data / Parameter | $EG_y$ |
| Data unit | GWh |

[19] SD_Report_2020_EN_Final.pdf (banpupower.com)

[20] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

| | |
|---|---|
| Description | Net electricity generated and delivered to the grid by all power sources serving the system, including low-cost/must-run power plants/units, in year y. |
| Source of data | - The data of the electricity generated and delivered to Lao PDR National Power Grid are from Electricity Statistic (2020), EDL<br>- The data of the electricity generated and delivered to Thailand National Power Grid are from Annual Report (2020& 2019), EGAT<br>The data of the electricity generated and delivered by Lao IPP directly supply to Thailand are from Electricity Statistic (2020),EDL |
| Value applied | See the Table 3 in Appendix 1. |
| Justification of choice of data or description of measurement methods and procedures applied | Data used are from Thailand authority, EGAT and Lao authority, EDL. |
| Purpose of Data | Calculation of baseline emissions |
| Comments | This parameter is fixed ex-ante for the entire crediting period |

| | |
|---|---|
| Data / Parameter | $EG_{import,y}$ |
| Data unit | MWh |
| Description | The electricity (MWh) imported from Malaysia, China and Vietnam Power Grid in year y. |
| Source of data | - Annual report (2020, 2019, 2018), EGAT<br>- Electricity Statistic (2020, 2019, 2018,), EDL |
| Value applied | See the Table 4 in Appendix 1 |
| Justification of choice of data or description of measurement methods and procedures applied | Data used are from Thailand authority, EGAT and Lao authority, EDL. |
| Purpose of Data | Calculation of baseline emissions |
| Comments | This parameter is fixed ex-ante for the entire crediting period |
| Data / Parameter | $EF_{grid,OM,y}$ |
| Data unit | $tCO_2/MWh$ |
| Description | Operating Margin $CO_2$ emission factor in year y |

| | |
|---|---|
| Source of data | Calculated. |
| Value applied | 0.5542 |
| Justification of choice of data or description of measurement methods and procedures applied | Calculated as per "Tool to calculate the emission factor for an electricity system, version 07" as 3-year generation weighted average using data for the years 2018, 2019 & 2020. |
| Purpose of Data | Baseline Emission calculation |
| Comments | This parameter is fixed ex-ante for the entire crediting period |

| | |
|---|---|
| Data / Parameter | $EF_{grid,BM,y}$ |
| Data unit | $tCO_2/MWh$ |
| Description | Build Margin $CO_2$ emission factor in year y |
| Source of data | Calculated. |
| Value applied | 0 |
| Justification of choice of data or description of measurement methods and procedures applied | Calculated as per Tool 07 : "Tool to calculate the emission factor for anelectricity system, version 07" |
| Purpose of Data | Baseline Emission calculation |
| Comments | This parameter is fixed ex-ante for the entire crediting period |

| | |
|---|---|
| Data / Parameter | $EF_{grid,CM,y}$ |
| Data unit | $tCO_2/MWh$ |
| Description | Combined Margin CO2 emission factor in year y |
| Source of data | Calculated |
| Value applied | 0.5542 |
| Justification of choice of data or description of measurement methods and procedures applied | The combined margin emissions factor is calculated as<br>follows: $EF_{grid,CM,y}= EF_{grid,OM,y}* WOM + EF_{grid.BM,y}* W_{BM}$<br>Where: |

| | |
|---|---|
| | $EF_{grid,BM,y}$= Build margin $CO_2$ emission factor in year y($tCO_2/MWh$)<br><br>$EF_{grid,OM,y}$= Operating margin $CO_2$ emission factor in year y<br><br>($tCO_2/MWh$)<br><br>$W_{OM}$ = Weighting of operating margin emissions factor (%) =100% (for simplified CM)<br><br>$W_{BM}$= Weighting of build margin emissions factor (%) = 0% (for simplified CM) |
| Purpose of Data | Baseline Emission calculation |
| Comments | This parameter is fixed ex-ante for the entire crediting period |

| | |
|---|---|
| Data / Parameter | $Cap_{PJ}$ |
| Data unit | MW |
| Description | Installed capacity of the hydro power plant after the implementation of the project activity |
| Source of data | Project site |
| Value applied | 10.5 |
| Justification of choice of data or description of measurement methods and procedures applied | Use the data in the FSR at start of the project. Measure by check the nameplate after operation |
| Purpose of Data | Project emission |
| Comments | - |

## 5.2 Data and Parameters Monitored

| | |
|---|---|
| Data / Parameter | $EG_{PJ,y}$ |
| Data unit | MWh/y |
| Description | Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh |

| | |
|---|---|
| Source of data | Monthly joint meter reading reports |
| Description of measurement methods and procedures to be applied | The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations. |
| Frequency of monitoring/recording | Continuous measurement & Monthly recording |
| Value applied | 44,270 |
| Monitoring equipment | The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. |
| QA/QC procedures to be applied | Lao electric state enterprise will procure and install the main and backup meter *at* the project proponent substation. The frequency of calibration is annually. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate Guidelines will be applied appropriately to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration frequency is under control of state electricity board and PP does not have any control on it. PP is getting value of net electricity supplied to grid and the same is considered the monitoring parameter. The billing is raised based on substation meters. |
| Purpose of data | Calculation of baseline emissions |
| Calculation method | Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh) |
| Comments | Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VERs for this project activity, whichever occurs later. |

## 5.3 Monitoring Plan

The purpose of the monitoring plan is to ensure that the monitoring and calculation of emission reductions of the project within the crediting period is complete, consistent, clear and accurate. The plan will be implemented by the project owner with the support of the grid corporation.

1. Monitoring organization

The monitoring process will be carried out and responsibility by the project owner. A monitoring panel will be established by the plant managers to be in charge of monitoring the data and information relating to the calculation of emission reductions with the cooperation of the Technical and Financial Department. A VCS manager will be assigned full charge the monitoring works. The operation and management structure are shown below:

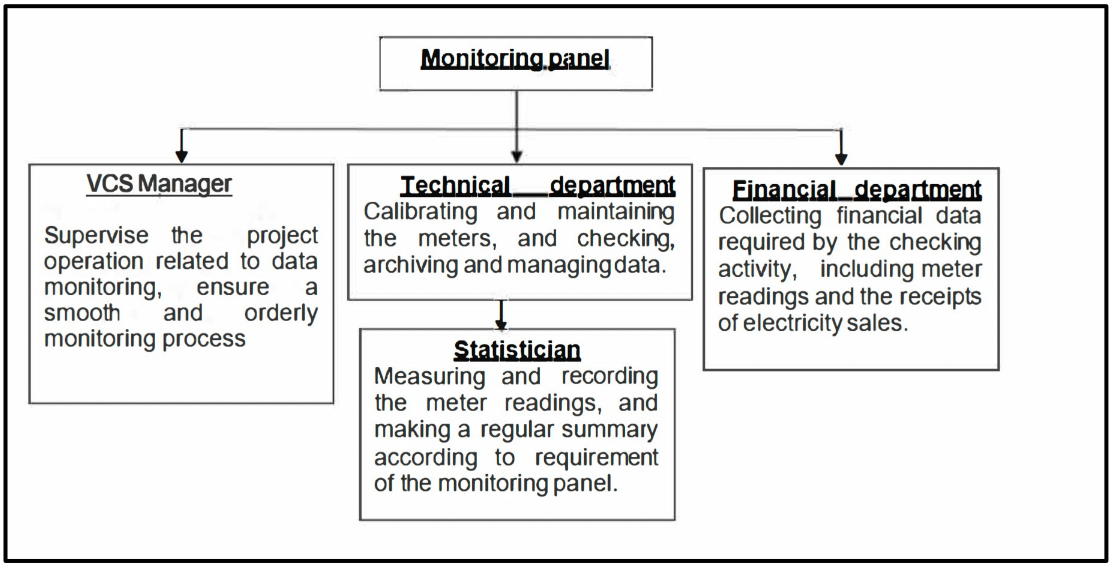


Fig. 3 Organization structure of the monitoring activity

2. Monitoring apparatus and installation

The meter(s) will be installed at the project site, to monitoring the input/output electricity at the grid side. The meter(s) will be installed in accordance with relevant national or international standard. The serial numbers of meters will be provided once the project is commissioned.

3. Data collection:

The specific steps for data collection and reporting are listed below:

a) During the crediting period, both the Lao electric state enterprise (grid company) and the project proponent will record the values displayed by the main meter.

b) Simultaneously to step a), the project owner will both record the values displayed by the backup meters.

c) The meters will be calibrated according to the relevant regulation and request of Project proponent (PP)

d) The main meter's readings will be cross-checked with record document confirmed by PP.

e) The PP and the Lao electric state enterprise will record both output and input power readings from the main meter. These data will be used to calculate the amount of net electricity delivered to the grid.

f) The project owner will be responsible of providing copies of record document confirmed by Lao electric state enterprise to the DOE for verification.

If the reading of the main meter in a certain month is inaccurate and beyond the allowable error or the meter doesn't work normally, the grid-connected power generation shall be determined by following measures:

g) Read the data of the backup meters.

h) If the backup meter's data is not so accurate as to be accepted, or the practice is not standardized, the project owner and the grid corporation should jointly make a reasonable and conservative estimation method which can be supported by sufficient evidence and proved to be reasonable and conservative when verified by DOE.

i) If the project proponent and the grid corporation don't agree on an estimated method, arbitration will be conducted according the procedures set by the agreement to work out an estimation method.

4. Calibration

The calibration frequency of the monitoring meters will be annually. The accuracy of the monitoring meters will not less than 0.2s. Calibration of Meters should be implemented according to relevant standards and rules accepted by the Lao electric state enterprise. After the examination, the meters should be sealed. The lift of the seals requires the presence of both the project owner and the grid company. One party must not lift the seals or fiddle with the meters without the presence of the other party.

All the meters installed shall be tested by a qualified metering verification institution commissioned jointly by the project owner and the grid company within 10 days after:

1) Detection of a difference larger than the allowable error in the readings of both meters;

2) The repair of all or part of meter caused by the failure of one or more parts to operate in accordance with the specifications.

5. Data management system

Physical document such as the plant electrical wiring diagram will be gathered with this monitoring plan in a single place. In order to facilitate auditors" access to project documents, the project materials and monitoring results will be indexed. All paper-based information will be stored by the technical department of the project owner and all the material will have a copy for backup. All data, including calibration records, will be kept until 2 years after the end of the total crediting period.

6. Monitoring Report

During the crediting period, at the end of each year, the monitoring officer shall produce a monitoring report covering the past monitoring period. The report shall be transmitted to the General Manager who will check the data and issue a final monitoring report in the name of the project participants. Once the final report is issued, it will be submitted to the DOE for verification

# APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information has been excluded from the public version of the project description.

# APPENDIX 2: BASELINE INFORMATION

Table 1: Amount of fossil fuel type i consumed in the project electricity system in year y ($FC_{i,y}$)

| | Unit | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Lao PDR | tonnes | 20,111,689.80 | 18,435,902.60 | 18,469,552.94 |
| Thailand | tonnes | 48,641,000 | 50,831,000 | 48,019,000 |

Table 2: The $CO_2$ emission factor per unit of fuel i in year y

| | Unit | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Lao PDR | $tCO_{2e}$ | 10,054,839 | 9,217,030 | 9,233,853 |
| Thailand | $tCO_{2e}$ | 103,051,919 | 105,981,364 | 100,764,172 |

Table 3 : Net electricity generated and delivered to the grid by all power sources serving the system, including low-cost/must-run powerplants/units, in year y ($EG_y$)

| | Unit | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Lao PDR | GWh | 9,386.77 | 8,166.26 | 11,442.47 |
| Thailand | GWh | 164,872.12 | 172,347.52 | 162,373.45 |

Table 4 : The electricity imported from the connected system( $EG_{import,y}$)

| | Unit | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Lao imports from Vietnam | GWh | 26.32 | 25.58 | 26.34 |
| Lao imports from China | GWh | 48.08 | 14.24 | 95.98 |
| Thailand imports from Lao (Lao IPP directly supply to Thailand) | GWh | 26386.04 | 25407.09 | 29446.64 |
| Thailand imports from Malaysia | GWh | 119.57 | 118.39 | 126.05 |